Finite-Time Tracking of Arbitrary Swarm Formations with Safety and Connectivity Certificates

Yash Purswani*, Shivansh Agarwal* and Anuj Tiwari

***Abstract*— Certifying finite arrival times and convergence rates is essential for coordinating multi-agent systems in time-critical missions. However, convergence to a desired formation does not, by itself, guarantee collision avoidance or communication connectivity during transients. Existing swarm models can achieve finite-time tracking of collective variables, but may not provide independent settling guarantees for agents at arbitrarily prescribed stations. This paper proposes a first-order control framework for tracking arbitrary, time-varying formation stations relative to a moving reference. A component-wise fractional-power feedback ensures exact finite-time station tracking with an explicit settling-time bound independent of the interaction graph. To address transient safety, we derive a graph-dependent exponential bound on relative disagreement that remains valid for time-varying and switching graphs, including disconnected modes. Combining this bound with the geometry of the prescribed stations yields sufficient conditions for pairwise collision avoidance and preservation of a connected spanning communication backbone. These conditions further provide a coupling-gain design rule based on the available geometric clearance throughout the maneuver. Numerical simulations illustrate finite-time tracking, transient formation deformation, and the effectiveness and conservatism of the proposed safety and connectivity certificates.**

***Index Terms*— Multi-agent systems, formation control, finite-time stability, collision avoidance, network connectivity.**

I. Introduction

Multi-agent systems enable coordinated tasks in which individual robots must move collectively while maintaining a prescribed spatial arrangement [1]. In many applications, the desired formation is asymmetric or time-varying, and each agent must reach its assigned station relative to a moving reference. For time-critical missions, asymptotic convergence may be insufficient: all agents must reach their stations within a finite time, without colliding or losing essential communication links during the transient. These requirements are coupled, since tracking errors can cause collisions or push communicating agents beyond sensing range even when the desired formation is itself feasible.

Finite-time stability provides exact convergence within a finite settling time [2]. However, finite-time convergence alone does not certify that the transient remains physically feasible. Existing formation-control approaches address related objectives through different mechanisms. For example, prescribed-performance distance-based formation control shapes formation errors [3], while barrier-function methods have been used to jointly enforce collision avoidance and connectivity maintenance for constrained second-order agents [4]. The problem considered here is complementary: to obtain an explicit, graph-independent deadline for tracking individually prescribed stations and use a separate analytical bound to certify collision avoidance and persistence of essential communication links.

A particularly relevant starting point is the kinematic swarm model of Fedele and D'Alfonso [5], which achieves finite-time centroid tracking and finite-time aggregation around the centroid trajectory. Its interaction dynamics determine the resulting aggregation, however, rather than allowing each agent's station to be prescribed independently. Moreover, the centroid-level convergence result does not itself provide a finite-time settling guarantee for every agent's error relative to an arbitrary assigned station. These gaps motivate separating station tracking from the analysis of relative agent motion.

This paper develops a first-order controller for tracking arbitrary prescribed station trajectories. A component-wise fractional-power feedback yields an explicit finite-time settling bound independent of the interaction graph. A second, graph-dependent estimate bounds relative disagreement for undirected time-varying or switching graphs, including disconnected modes. Combining this estimate with the geometry of the desired stations gives sufficient conditions for collision avoidance and preservation of a designer-selected connected spanning communication backbone. The conditions further yield a coupling-gain inequality based on the available time-varying geometric clearance.

The main contributions are:

1) A first-order station-tracking controller with an explicit, graph-independent finite-time settling bound for arbitrary prescribed C^1 station trajectories.
2) A graph-dependent exponential disagreement bound valid for undirected time-varying or switching graphs, including disconnected modes.
3) Sufficient geometric conditions for pairwise collision avoidance and persistence of a connected spanning communication backbone throughout the transient.
4) A coupling-gain design inequality derived from the time-varying clearance margin, illustrated through numerical examples that distinguish formal certification from observed safe behavior.

II. Background and Problem Statement

Formation control is commonly formulated using position-, displacement-, or distance-based objectives, with the appropriate choice depending on the available measurements and

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the desired formation specification [1]. Finite-time stability provides exact convergence in finite time rather than merely asymptotic convergence [2]; however, finite-time convergence of a collective quantity does not automatically imply that every agent reaches an independently assigned, time-varying station by a common explicit deadline. For example, the kinematic swarm model of Fedele and D'Alfonso establishes finite-time centroid tracking and aggregation [5], whereas the present problem prescribes each station trajectory directly and seeks a graph-independent bound on the complete station-tracking error. Distance-based prescribed-performance methods shape formation errors [3], while other approaches jointly address formation navigation, collision avoidance, and connectivity under different dynamics and control constructions [4]. These results motivate distinguishing convergence to a formation from tracking prescribed stations while certifying transient feasibility.

Collision avoidance and communication connectivity have also been studied using potential functions, barrier-based constraints, and connectivity-preserving control laws. Such methods directly incorporate safety constraints into the controller or constrain the evolution of selected inter-agent distances. In contrast, this paper uses a bound on the disagreement component of station-tracking error to enclose actual pairwise distances around their prescribed values. This gives a complementary, sufficient certificate: the method does not replace barrier-based enforcement, and failure of its certificate does not imply that a trajectory is unsafe. The disagreement analysis is related to graph-Laplacian methods in consensus theory, where algebraic connectivity characterizes contraction rates and switching topologies require appropriate regularity assumptions [6]. The contribution here is to connect such a contraction estimate to explicit geometric clearance conditions for collision avoidance and persistence of a designer-selected spanning communication backbone, and then to derive a coupling-gain condition from the planned station trajectories.

a) Problem statement.: Consider n first-order agents with positions $x_i \in \mathbb{R}^d$ and control inputs u_i, and prescribed continuously differentiable station trajectories $q_i(t) = r(t) + p_i(t)$. Define the station-tracking errors $\delta_i = x_i - q_i$. Given a minimum allowed pairwise separation d_{safe}, an operational communication range R_c, and a designer-selected connected spanning backbone $\mathcal{H}$, the objectives are to: (i) design a distributed controller that guarantees exact finite-time convergence $\delta_i(t) = 0$ by an explicit deadline independent of the communication graph; (ii) derive verifiable sufficient conditions ensuring $\|x_i(t) - x_j(t)\| > d_{\text{safe}}$ for all pairs and all times, while every edge of $\mathcal{H}$ remains within R_c; and (iii) provide a coupling-gain selection rule based on the prescribed station geometry and initial disagreement. The safety and connectivity claims are conditional certificates, not claims that trajectories failing the certificate must violate a constraint.

III. Model and Control Law

Consider $n \geq 2$ agents with first-order dynamics

$$\dot{x}_i = u_i, \qquad i = 1, \ldots, n, \tag{1}$$

where $x_i \in \mathbb{R}^d$ and $u_i \in \mathbb{R}^d$ denote the position and control input of agent i, respectively. To achieve finite-time tracking of arbitrary time-varying formations, we consider the distributed controller

$$u_i = \dot{q}_i - \delta_i - \Phi_\beta(\delta_i) - \sum_{j=1}^{n} w_{ij} M(\delta_i - \delta_j), \tag{2}$$

where q_i is the desired station of agent i, $\delta_i = x_i - q_i$ is its tracking error, Φ_β is a component-wise fractional-power feedback, w_{ij} are nonnegative interaction weights, and $M \in \mathbb{R}^{d\times d}$ is a symmetric positive semidefinite coupling matrix.

The desired station trajectories are defined as

$$q_i(t) = r(t) + p_i(t), \qquad \delta_i(t) = x_i(t) - q_i(t), \tag{3}$$

where $r(t) \in \mathbb{R}^d$ is a reference trajectory and $p_i(t) \in \mathbb{R}^d$ is the desired offset of agent i. Unless stated otherwise, the station trajectories q_i are continuously differentiable with locally bounded derivatives. This formulation allows the desired formation to translate and change shape over time. For $0 < \beta < 1$, the component-wise fractional-power feedback is defined by

$$[\Phi_\beta(z)]_k = \operatorname{sign}(z_k)|z_k|^\beta, \qquad k = 1, \ldots, d, \tag{4}$$

with $\Phi_\beta(0) = 0$.

The interactions among agents are represented by an undirected weighted graph with symmetric weights $w_{ij} = w_{ji} \geq 0$ and $w_{ii} = 0$. Let $W = [w_{ij}]$ denote the weighted adjacency matrix and define the weighted graph Laplacian as

$$L_w = \operatorname{diag}\left(\sum_{j=1}^{n} w_{ij}\right) - W. \tag{5}$$

For the range-dependent interaction model considered in this work, the weights are given by

$$w_{ij}(x,t) = \frac{a_{ij}(x,t)}{\|x_i - x_j\|^\nu + \varepsilon_w}, \qquad \nu \geq 1, \quad \varepsilon_w > 0, \tag{6}$$

where the communication indicator is determined strictly by the operational range,

$$a_{ij}(x) = \begin{cases} 1, & \|x_i - x_j\| \leq R_c, \\ 0, & \|x_i - x_j\| > R_c, \end{cases} \qquad a_{ij} = a_{ji}, \quad a_{ii} = 0. \tag{7}$$

Thus, the active communication graph is the range graph induced by the current agent positions; R_c is the operational communication radius. The positive constant ε_w prevents the weights from becoming singular when agents coincide. The designer also specifies a connected spanning backbone $\mathcal{H}$ (at minimum, a spanning tree) whose edges must remain active throughout the maneuver.

Using the station-error definition in (3), we have

$$\delta_i - \delta_j = (x_i - x_j) - (q_i - q_j). \tag{8}$$

Thus, the coupling term can be computed from relative agent positions and the desired relative station assignments. Substituting the controller (2) into the agent dynamics (1) yields the closed-loop error dynamics

$$\dot{\delta} = -\delta - \Phi_\beta(\delta) - (L_w(x,t) \otimes M)\delta, \tag{9}$$

where $\delta = [\delta_1^\top, \dots, \delta_n^\top]^\top$ and $\Phi_\beta(\delta)$ denotes the stacked component-wise fractional-power feedback.

Assumption 1. *The range-induced weights are locally bounded between topology switches. The range graph is undirected, its switching rule is well posed, and it has finitely many switches on every bounded time interval; agent states remain continuous at switching times. Each agent knows its own state, reference and station trajectory (including $\dot{q}_i$), and exchanges its state and station information with currently active neighbors. Hence, each agent can compute δ_i and the relative errors $\delta_i - \delta_j$ for its active neighbors. The selected backbone $\mathcal{H}$ is specified by the designer and is a connected spanning subgraph of the planned communication network.*

IV. Finite-Time Station Tracking and Reference Convergence

Let $\delta_0 = \delta(t_0)$ and define the non-negative envelope

$$\mathcal{R}(t) = \left[\left(1 + \|\delta_0\|^{1-\beta}\right)e^{-(1-\beta)(t-t_0)} - 1\right]^{1/(1-\beta)}, \tag{10}$$

and finite-time arrival deadline be

$$T = t_0 + \frac{\ln(1 + \|\delta_0\|^{1-\beta})}{1-\beta}. \tag{11}$$

Theorem 1 (Finite-time assigned-station tracking)**.** *Under Assumption 1, with $M = M^\top \succeq 0$ and $0 < \beta < 1$, every solution of* (9) *satisfies*

$$\|\delta(t)\| \le \mathcal{R}(t), \qquad t \ge t_0, \tag{12}$$

and $\delta(t) = 0$ for all $t \ge T$. Thus, $x_i(t) = q_i(t)$ for all agents and all $t \ge T$. The deadline is independent of the interaction graph, its connectivity, and M.

Proof. Consider $V = \frac{1}{2}\|\delta\|^2$. Symmetry of the weights and positive semi-definiteness of M give the edge-energy identity

$$\begin{aligned}\mathcal{D} &:= \delta^\top (L_w \otimes M)\delta \\ &= \frac{1}{2}\sum_{i,j} w_{ij}(\delta_i - \delta_j)^\top M(\delta_i - \delta_j) \ge 0.\end{aligned} \tag{13}$$

Since $1+\beta < 2$, norm monotonicity implies $\sum_{\ell=1}^{nd} |\delta_\ell|^{1+\beta} \ge \|\delta\|_2^{1+\beta}$. Consequently along (9),

$$\dot{V} = -\|\delta\|^2 - \sum_{i=1}^{n}\sum_{k=1}^{d} |\delta_{ik}|^{1+\beta} - \mathcal{D} \tag{14}$$

$$\le -\|\delta\|^2 - \|\delta\|^{1+\beta} = -2V - (2V)^{(1+\beta)/2}, \tag{15}$$

Let $z = \|\delta\| = \sqrt{2V}$. Whenever $z > 0$, (15) implies $\dot{z} \le -z - z^\beta$. With $y = z^{1-\beta}$,

$$\dot{y} \le -(1-\beta)(y+1). \tag{16}$$

Integration yields $y(t) + 1 \le (y(t_0)+1)e^{-(1-\beta)(t-t_0)}$ until the tracking error reaches zero. This comparison bound establishes the bounds in (10) and (11), guaranteeing that the error reaches zero within a finite time. Since the origin is invariant under the error dynamics, the error remains zero after reaching it. Since V is non-increasing, the error remains bounded; the regularity and well-posed switching assumptions ensure the comparison holds across topology changes. □

A. Centroid and Formation Guarantees

Assume the station offsets are centered:

$$\sum_{i=1}^{n} p_i(t) = 0, \qquad \forall t \ge t_0. \tag{17}$$

Let $c(t) = \frac{1}{n}\sum_i x_i(t)$ be the swarm centroid. Since $q_i = r + p_i$, the mean desired station is r.

Corollary 1 (Reference tracking)**.** *Under the centering condition* (17)*,*

$$\|c(t) - r(t)\| \le \frac{\mathcal{R}(t)}{\sqrt{n}}, \qquad t \ge t_0. \tag{18}$$

Consequently, $c(t) = r(t)$ for all $t \ge T$.

Proof. Since $q_i = r + p_i$ and $\sum_{i=1}^n p_i = 0$, the centroid error satisfies

$$c - r = \frac{1}{n}\sum_{i=1}^{n}(x_i - q_i) = \frac{1}{n}\sum_{i=1}^{n}\delta_i.$$

By the Cauchy–Schwarz inequality,

$$\|c - r\| \le \frac{1}{\sqrt{n}}\left(\sum_{i=1}^{n}\|\delta_i\|^2\right)^{1/2} = \frac{\|\delta\|}{\sqrt{n}}.$$

Applying the bound $\|\delta(t)\| \le \mathcal{R}(t)$ from Theorem 1 gives the stated inequality. Since $\mathcal{R}(t) = 0$ for all $t \ge T$, it follows that $c(t) = r(t)$ for all $t \ge T$. □

Remark 1. *The finite-time reference result is inherited from exact station tracking. If the offsets are not centered, the station theorem remains valid but the swarm centroid tracks the mean station $r + \frac{1}{n}\sum_i p_i$, not r.*

V. Disagreement-Based Safety and Connectivity Certificates

The complete station error contains both a common component and a disagreement component. Only disagreement changes relative positions. Define

$$\begin{aligned}&\Pi = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad \eta = (\Pi \otimes I_d)\delta, \\ &\mu = \lambda_{\min}(M) \ge 0.\end{aligned} \tag{19}$$

Thus $\eta_i = \delta_i - \bar{\delta}$ and $\delta_i - \delta_j = \eta_i - \eta_j$.

Proposition 1 (Graph-dependent disagreement contraction)**.** *For the undirected weighted graph in* (6)*, every solution satisfies*

$$\|\eta(t)\| \le \|\eta(t_0)\| \exp\left[-\int_{t_0}^{t}\left(1 + \mu\lambda_2(L_w(s))\right)\,ds\right] \tag{20}$$

Here $\lambda_2(L_w)$ is the algebraic connectivity, with $\lambda_2 = 0$ for a disconnected graph and $\mu = \lambda_{\min}(M)$.The bound remains valid through admissible topology switches.

Proof. Since $\Pi L_w = L_w$ and $\Pi\eta = \eta$, the disagreement dynamics followed from (9) are

$$\dot{\eta} = -\eta - (\Pi \otimes I_d)\Phi_\beta(\delta) - (L_w \otimes M)\eta. \tag{21}$$

For $W_\eta = \frac{1}{2}\|\eta\|^2$, use $\eta^\mathsf{T}(\Pi \otimes I_d) = \eta^\mathsf{T}$ to obtain

$$\dot{W}_\eta = -\|\eta\|^2 - \eta^\mathsf{T}\Phi_\beta(\delta) - \eta^\mathsf{T}(L_w \otimes M)\eta. \tag{22}$$

The non-linear term is non-negative because component-wise monotonicity of Φ_β implies

$$\eta^\mathsf{T}\Phi_\beta(\delta) = \frac{1}{2n}\sum_{i,j}(\delta_i - \delta_j)^\top\big(\Phi_\beta(\delta_i) - \Phi_\beta(\delta_j)\big) \geq 0 \tag{23}$$

Since $M \succeq 0$ and $\mu = \lambda_{\min}(M)$, the Rayleigh quotient bound gives

$$\eta^\top(L_w \otimes M)\eta \geq \mu\lambda_2(L_w)\|\eta\|^2$$

Discarding the non-positive term,

$$\dot{W}_\eta \leq -2\big[1 + \mu\lambda_2(L_w)\big]W_\eta.$$

Applying the comparison lemma and integrating yields (20). □

A. Pairwise Distance Bounds

Let $d_{ij}^q(t) = \|q_i(t) - q_j(t)\|$ and $d_{ij}^x(t) = \|x_i(t) - x_j(t)\|$.

Lemma 1 (Relative-distance envelope)**.** *For every pair i, j,*

$$d_{ij}^q(t) - \sqrt{2}\,\|\eta(t)\| \leq d_{ij}^x(t) \leq d_{ij}^q(t) + \sqrt{2}\,\|\eta(t)\|. \tag{24}$$

Proof. Since $x_i - x_j = (q_i - q_j) + (\eta_i - \eta_j)$, the triangle inequality gives

$$\|x_i - x_j\| \leq \|q_i - q_j\| + \|\eta_i - \eta_j\|,$$

while the reverse-triangle inequality yields

$$\|x_i - x_j\| \geq \|q_i - q_j\| - \|\eta_i - \eta_j\|.$$

Moreover,

$$\|\eta_i - \eta_j\| \leq \|\eta_i\| + \|\eta_j\| \leq \sqrt{2}(\|\eta_i\|^2 + \|\eta_j\|^2) \leq \sqrt{2}\,\|\eta\|.$$

Hence, the two inequalities above establishes the claimed bounds. □

To preserve communication links that are critical for coordination during formation maneuvers, we consider a designer-selected connected spanning backbone $\mathcal{H} = (\mathcal{V}, \mathcal{E}_\mathcal{H})$. Let $L_\mathcal{H}$ denote the unweighted Laplacian of $\mathcal{H}$. We define the station-edge length envelope and the minimum pairwise station distance as

$$\begin{aligned} \ell_\mathcal{H}(t) &= \max_{(i,j)\in\mathcal{E}_\mathcal{H}} \|q_i(t) - q_j(t)\|, \\ d_q(t) &= \min_{i<j} \|q_i(t) - q_j(t)\|. \end{aligned} \tag{25}$$

Let $\lambda_2(L_\mathcal{H}) > 0$ and let $\underline{w} > 0$ be a lower bound on every active backbone-edge weight. For the range-dependent weights in (6), one may choose $\underline{w} = (R_c^\nu + \varepsilon_w)^{-1}$. Define

$$\begin{aligned} \gamma_\mathcal{H} &= 1 + \mu\underline{w}\lambda_2(L_\mathcal{H}), \\ \rho_0 &= \sqrt{2}\,\|\eta(t_0)\|. \end{aligned} \tag{26}$$

Since $\eta = (\Pi \otimes I_d)\delta$ and Π is an orthogonal projection, it cannot increase the Euclidean norm; hence $\|\eta(t)\| \leq \|\delta(t)\|$. By Theorem 1, this gives $\sqrt{2}\|\eta(t)\| \leq \sqrt{2}\,\mathcal{R}(t)$. Moreover, Proposition 1 gives exponential contraction of the disagreement. While all backbone edges remain active, $L_w \succeq \underline{w}L_\mathcal{H}$, so the contraction rate is at least $\gamma_\mathcal{H}$. Consequently,

$$\rho_\mathcal{H}(t) = \min\left\{\sqrt{2}\,\mathcal{R}(t),\, \rho_0 e^{-\gamma_\mathcal{H}(t-t_0)}\right\}. \tag{27}$$

This is an a priori envelope on the maximum pairwise disagreement for every pair of agents, provided the backbone remains available.

Theorem 2 (Joint collision avoidance and connectivity preservation)**.** *If the planned stations satisfy, for all $t \geq t_0$,*

$$d_q(t) > d_{\text{safe}} + \rho_\mathcal{H}(t), \tag{28}$$

$$\ell_\mathcal{H}(t) < R_c - \rho_\mathcal{H}(t), \tag{29}$$

then every pair of agents remains strictly separated by more than d_{safe}, every backbone edge remains within range R_c, and the communication graph remains connected. The station-tracking deadline T in (11) *is unchanged.*

Proof. At t_0, (27) gives $\rho_\mathcal{H}(t_0) = \rho_0$, since $\|\eta(t_0)\| \leq \|\delta(t_0)\|$. By Lemma 1 and (28), all actual pairwise distances exceed d_{safe} initially. The same lemma and (29) put all backbone links strictly inside R_c initially.

Suppose a backbone edge first reaches R_c at time $t_e > t_0$. Until t_e, all backbone edges remain active, so the lower bound $L_w \succeq \underline{w}L_\mathcal{H}$ and Proposition 1 imply (27). By continuity, the bound also holds at t_e. For every backbone edge, Lemma 1 then gives $\|x_i(t_e) - x_j(t_e)\| \leq \ell_\mathcal{H}(t_e) + \rho_\mathcal{H}(t_e) < R_c$, a contradiction. Thus the backbone persists for all time. Finally, the lower half of Lemma 1 and (28) imply $d_{ij}^x(t) > d_{\text{safe}}$ for every pair and all times. The station-tracking theorem is unaffected by the graph argument. □

B. Coupling-Gain Design Rule

Define the available geometric margin

$$s(t) = \min\{d_q(t) - d_{\text{safe}},\, R_c - \ell_\mathcal{H}(t)\}. \tag{30}$$

The margin depends only on the prescribed station trajectories, the selected safety threshold, the operational communication range, and the chosen backbone. It can therefore be computed offline, before execution, without live agent-state measurements. Geometric feasibility requires $s(t) > 0$ throughout the interval of interest. Suppose $s(t) > 0$ on $[t_0, T]$ and $0 < \rho_0 < s(t_0)$. The latter is an initial-margin condition: if it fails, increasing the coupling gain cannot make this particular certificate applicable. Define

$$\gamma_{\text{req}} = \sup_{t_0<t<T} \frac{\ln\big(\rho_0/s(t)\big)}{t - t_0}. \tag{31}$$

Corollary 2 (Sufficient coupling selection)**.** *A sufficient condition for Theorem 2 is*

$$1 + \lambda_{\min}(M)\,\underline{w}\,\lambda_2(L_{\mathcal{H}}) > \gamma_{\mathrm{req}}. \tag{32}$$

For $M = mI_d$, if $\gamma_{\mathrm{req}} > 1$, it is sufficient to choose

$$m > \frac{\gamma_{\mathrm{req}} - 1}{\underline{w}\lambda_2(L_{\mathcal{H}})}. \tag{33}$$

Thus, once the station plan and physical limits (d_{safe}, R_c) are specified, the margin and required rate can be computed offline and used to select a gain satisfying the certificate. This is a sufficient design rule, not an optimal-gain result; the selected gain should also be assessed against practical actuator limits, which are not modelled here.

Proof. By Theorem 2, it suffices to ensure that $\rho_{\mathcal{H}}(t) < s(t)$ throughout the maneuver. By the definition of γ_{req} in (31), $\gamma_{\mathcal{H}} > \gamma_{\mathrm{req}}$ ensures that

$$\rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)} < s(t), \qquad t \in (t_0, T).$$

Since $\rho_{\mathcal{H}}(t) \le \rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)}$ by (27), the sufficient conditions hold on (t_0, T). The strict initial condition handles $t = t_0$, while at $t = T$, $\rho_{\mathcal{H}}(T) = 0 < s(T)$. Thus, the certificate holds throughout the maneuver. In particular, its minimum margin

$$\Delta_{\min} := \min_{t\in[t_0,T]} [s(t) - \rho_{\mathcal{H}}(t)]$$

is strictly positive, quantifying the available clearance of the certificate. □

Remark 2. *If $\gamma_{\mathrm{req}} < 1$, the baseline local term already satisfies the rate condition with $M = 0$. The rule is sufficient, not necessary or optimal: failure of the inequality means that this certificate cannot conclude safety, not that a collision must occur. Increasing coupling cannot repair a failed initial-margin test $\rho_0 < s(t_0)$.*

VI. Numerical Validation

We evaluate the theoretical results through simulations of $n = 10$ agents in $\mathbb{R}^2$ using the benchmark parameters in Table I. The complete simulation framework and experiments are available at `https://github.com/yash-purswani/finite-time-swarm-control`.

Three scenarios are considered: (i) finite-time tracking and coupling-gain trade-offs, (ii) aggressive star compression and expansion to test transient safety and connectivity, and (iii) narrow-corridor traversal to demonstrate shape morphing. Unless stated otherwise, initial perturbations satisfy $\rho_0 < s(0)$. The star formation rotates at rate ω and scales as $a(t) = a_\infty + (a_0 - a_\infty)e^{-\kappa t}$. Connectivity is assessed using a 15-edge spanning backbone with $\lambda_2(L_{\mathcal{H}}) \approx 0.9316$, yielding the certified disagreement decay rate $\gamma_{\mathcal{H}} = 1 + 0.1540m$.

TABLE I: Benchmark simulation parameters.

Param.	Value	Description
n	10	Swarm size in $\mathbb{R}^2$
β	0.5	Fractional-power velocity exponent
d_{safe}	1.0 m	Minimum pairwise collision threshold
R_c	6.0 m	Maximum communication sensing radius
ε_w	0.05	Proximity weight regularizer
ν	1	Distance exponent in the interaction weights
$\underline{w}$	0.1653 m^{-1}	Uniform lower bound $1/(R_c + \varepsilon_w)$
$r(t)$	$[t,\ 2\sin(0.5t)]^{\mathsf{T}}$	Reference centroid trajectory
ω	$\pi/4$ rad/s	Star template angular rotation rate
M	mI_2	Cooperative coupling matrix ($m \ge 0$)

TABLE II: Settling times and control metrics across coupling gains m under rotating star formation ($\kappa = 2.0$).

m	$t_{10^{-6}}$ (s)	T (s)	Peak $\|u\|_2$	J_u
0.0	1.1517	1.3621	16.9828	429.44
0.5	0.8117	1.3621	17.0860	429.92
2.0	0.5250	1.3621	18.0322	431.55
5.0	0.4467	1.3621	22.2186	434.95
10.0	0.3933	1.3621	32.8156	440.74

A. Finite-Time Tracking and Coupling-Gain Trade-offs

We evaluate the influence of coupling gain m during rotating star compression ($\kappa = 2.0$, $a_0 = 1.70$ to $a_\infty = 0.72$). As shown in Figure 1 and Table II, the tracking error reaches zero by the common theoretical deadline, independent of the interaction topology and coupling strength. The reported numerical settling time $t_{10^{-6}}$ is the first time at which the stacked Euclidean error satisfies $\|\delta(t)\|_2 \le 10^{-6}$; it is a tolerance-based numerical metric, not a numerical verification of exact equality. Stronger coupling accelerates disagreement decay. The theoretical deadline remains unchanged, while the finite-tolerance settling time varies with m.

B. Transient Safety and Connectivity

We stress-test the certificate under rapid star compression ($\kappa = 5.0$) and expansion ($\kappa = 8.0$), comparing gains that satisfy the sufficient design condition with an uncoupled baseline ($m = 0.0$, $\gamma_{\mathcal{H}} = 1.0$); see Table III and Figure 2. Failure to satisfy the certificate is not, by itself, evidence of a safety or connectivity violation; the certificate is sufficient and may be conservative.

- **Compression ($\kappa = 5.0$, $\gamma_{\mathrm{req}} = 4.4882$):** The star contracts from $a_0 = 1.70$ down to $a_\infty = 0.72$. For uncoupled agents ($m = 0.0$), tracking lag causes relative distortion to exceed nominal clearances, resulting in a physical collision breach ($d_{x,\min} < d_{\mathrm{safe}}$) despite collision-free nominal stations. In contrast, certified coupling ($m = 30.0$) ensures $s(t) > \rho_{\mathcal{H}}(t)$ and guarantees collision avoidance ($d_{x,\min} > d_{\mathrm{safe}}$).
- **Expansion ($\kappa = 8.0$, $\gamma_{\mathrm{req}} = 4.1367$):** The star expands from $a_0 = 1.70$ up to $a_\infty = 3.02$, leaving a narrow terminal clearance margin of 0.112 m along

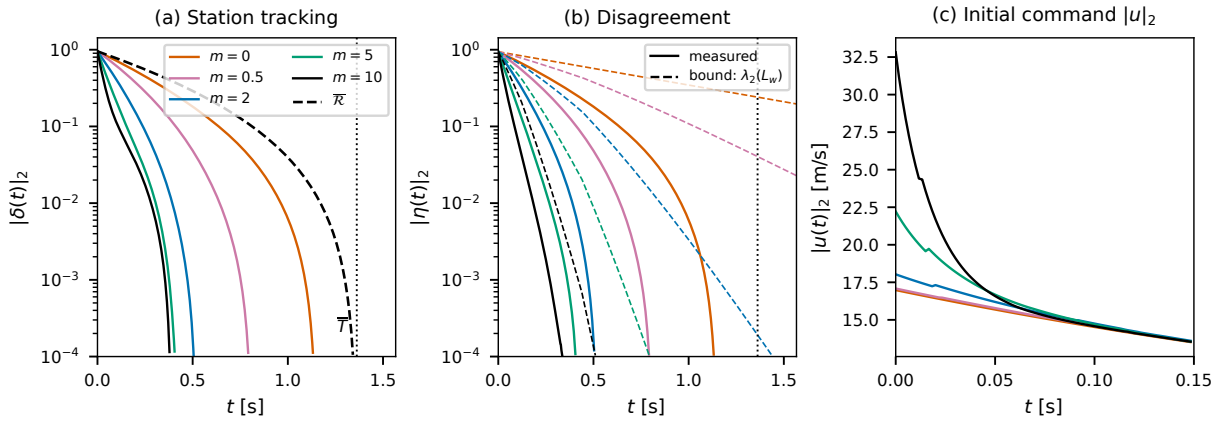


Fig. 1: Finite-time tracking, disagreement decay, and initial control effort across coupling gains m ($\kappa = 2.0$). (a) Station tracking $\|\delta(t)\|_2$ against theoretical bound $\overline{\mathcal{R}}(t)$ and deadline T. (b) Disagreement $\|\eta(t)\|_2$ against analytical envelope bounds. (c) Transient control command $\|u(t)\|_2$ over $t \in [0, 0.15]$ s.

the backbone. Uncoupled tracking suffers outward lag, breaching communication range ($\ell_{x,\max} > R_c$) for over 0.45 s. The selected gain $m = 30.0$ satisfies the certificate ($\Delta_{\min} > 0$) and the simulated backbone links remain below R_c.

C. Arbitrary Formation Morphing and Constrained Corridor Traversal

The final scenario combines time-varying station tracking, geometric feasibility, collision avoidance, and backbone connectivity in a corridor maneuver. The swarm traverses a corridor ($x \in [12, 30]$ m, half-width $y_{\text{wall}} = 2$ m) by smoothly blending, using C^2 quintic polynomials, from a circular ring into a slender chevron and back to a ring after exiting (Figure 3). Connectivity is assessed on a designer-selected 10-cycle backbone $\mathcal{H}$. For this planned maneuver, the initial-margin and geometric-feasibility conditions are satisfied, and the certificate remains satisfied over the maneuver. The agents reach their stations by the finite-time deadline ($T \approx 1.14$ s); the observed minimum separation is at least 1.32 m and the maximum backbone-edge length is at most 2.22 m. This example therefore illustrates the combined use of the tracking controller and the safety/connectivity certificate, rather than only an empirical corridor traversal.

VII. Conclusion

This paper presented a distributed first-order controller that tracks arbitrary C^1 station trajectories in finite time with a graph-independent deadline. A separate disagreement estimate bounds deviations of actual pairwise distances from the prescribed station geometry. For a designer-selected connected spanning backbone, this bound yields sufficient conditions for collision avoidance and communication-range preservation, together with an offline coupling-gain rule based on the planned geometric margin and initial disagreement. The compression, expansion, and corridor scenarios illustrate both the usefulness and the conditional nature of the certificate: satisfying it guarantees the stated constraints under the assumptions, whereas failing it is inconclusive and does not by itself imply a violation. The analysis is based on ideal first-order kinematics and does not account for actuator saturation or other physical input constraints.

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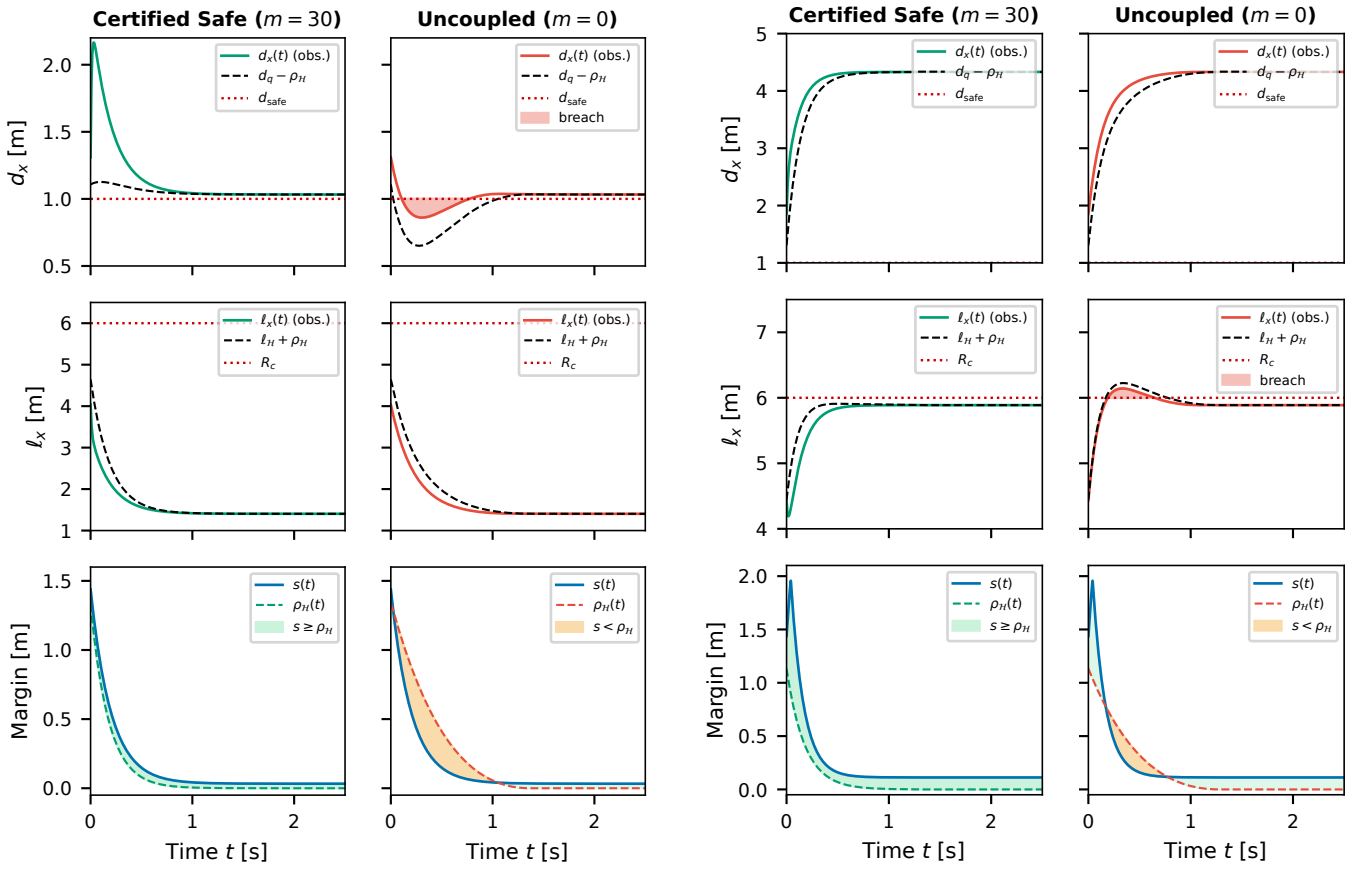
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(a) Formation compression ($\kappa = 5.0$).

(b) Formation Expansion ($\kappa = 8.0$).

Fig. 2: Transient safety and connectivity verification over $t \in [0, 2.5]$ s. In both subfigures, columns compare the certified safe design ($m = 30.0$, left) against the uncoupled baseline ($m = 0.0$, right). Rows display: minimum pairwise separation $d_x(t)$ (top), maximum pairwise distance $\ell_x(t)$ (middle), and geometric margin $s(t)$ versus disagreement tube $\rho_{\mathcal{H}}(t)$ (bottom). **(a) Left:** Star compression ($\kappa = 5.0$), where uncoupled tracking exhibits a physical collision breach ($d_x < d_{\text{safe}}$). **(b) Right:** Star expansion ($\kappa = 8.0$), where uncoupled tracking exhibits a transient communication-range violation ($\ell_x > R_c$). A failed certificate alone is inconclusive and does not imply a violation.

TABLE III: Certificate and observed transient-geometry outcomes for aggressive star maneuvers ($t \in [0, 2.5]$ s). A failed certificate is inconclusive rather than a prediction of failure.

Maneuver	Case	m	κ	$\gamma_{\mathcal{H}}$	γ_{req}	$t_{10^{-6}}$ (s)	$d_{x,\min}$ (m)	$\ell_{x,\max}$ (m)	$\Delta_{\min}$ (m)	Observed outcome
Compression	Certified	30.0	5.0	5.6194	4.4882	0.3800	1.0326	4.0338	0.0338	Safe and connected
	Uncertified	0.0	5.0	1.0000	4.4882	1.1517	0.8605	4.0338	≤ 0	Collision observed
Expansion	Certified	30.0	8.0	5.6194	4.1367	0.3767	1.8550	5.8880	0.0909	Safe and connected
	Uncertified	0.0	8.0	1.0000	4.1367	1.1333	1.8550	6.1424	≤ 0	Range violation observed

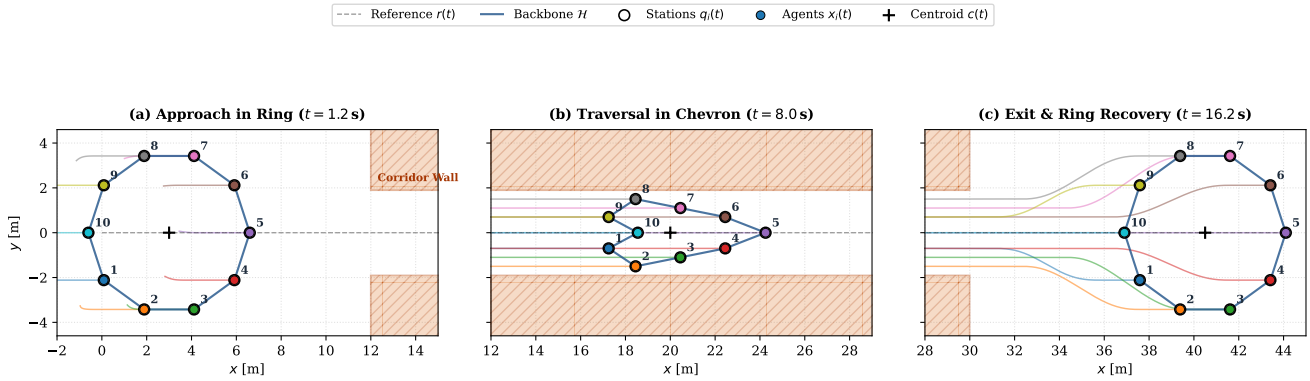


Fig. 3: Arbitrary formation morphing and constrained corridor traversal ($n = 10$, $m = 10.0$). Snapshots depict: (a) approach and finite-time convergence in ring geometry ($t = 1.2$ s), (b) morphing into a slender chevron to navigate the narrow corridor ($t = 8.0$ s), and (c) exit and ring recovery ($t = 16.2$ s). Target stations q_i (circles), agent positions x_i (colored dots), continuous trajectories, reference centroid $r(t)$ (dashed line), and the 10-cycle backbone $\mathcal{H}$ (solid blue edges) remain strictly collision-free ($d_{x,\min} \geq 1.32$ m) and connected ($\ell_{x,\max} \leq 2.22$ m).